Linear Algebraic Equations and Matrices (Ch 8)

Matrix notation: $[A]_{m \times n}$, a_{ij} : element of the matrix

Row vector : $[b] = [b_1 \quad b_2 \quad \cdots \quad b_n]$

Column vector : $[c] = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}$

Square matrix : $m = n$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

→ main diagonal (principal diagonal)

* System of linear equations: # equations (m) must be equal to # unknowns (n) for a unique sol^n to exist

Symmetric matrix : $a_{ij} = a_{ji}$

Diagonal " : off-diagonal elements are zero.

Identity matrix : diagonal matrix with all elements in the main diag. 1.

$$[A][I] = [A] = [I][A]$$

Upper triangular matrix :

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Lower Triang. Matr.

$$[A] = \begin{bmatrix} & & 0 \\ & & \\ & & \end{bmatrix}$$

Banded Matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & a_{43} & a_{44} \end{bmatrix}$$

→ has bandwidth = 3

tridiagonal matrix

Matrix Operating Rules

Equal $[A] = [B]$ iff $a_{ij} = b_{ij}$ $\forall$ i,j

Addition $[C] = [A] \pm [B]$, $c_{ij} = a_{ij} \pm b_{ij}$

Subtraction

Multiplication $[D] = g[A]$, $d_{ij} = g\, a_{ij}$

Product $[C] = [A][B]$, $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

$$[A]_{m \times n} \quad [B]_{n \times l} = [C]_{m \times l}$$

inner dimensions must be the same

Multiplication is **associative**:

$$([A][B])[C] = [A]([B][C])$$

Multiplication is **distributive**:

$$([A]+[B])[C] = [A][C] + [B][C]$$

Multiplication is not commutative:

$$[A][B] \neq [B][A]$$

Inverse: if $[A]$ is square and non-singular

$$[A]\underbrace{[A]^{-1}}_{\text{inv. of } A} = [I] = [A]^{-1}[A]$$

2x2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Transpose: $A^T \quad [A^T]_{ij} = a_{ji}$

Permutation Matrix (Transposition Matrix):

An identity matrix with rows and column interchanged

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$ switched row 1 and row 3 of an identity matrix

PA: will switch the corresponding rows of A.

$$PA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -7 & 4 \\ 8 & 3 & -6 \\ 5 & 1 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 9 \\ 8 & 3 & -6 \\ 2 & -7 & 4 \end{bmatrix}$$

Augmentation: add columns

- useful when performing same operation on the row of two matrix

$$\left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 0 \end{array}\right]$$

Representing Linear Algebraic Eq^{n}s in Matrix Form

3x3 set of linear eq^{n}s:

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}\right\}$$

$\underline{\underline{A}}\,\underline{x} = \underline{b}$ ← matrix, → vector

{ 3 eq^{n}s, 3 unknown

$$\underline{\underline{A}} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

To solve for $\underline{x}$

$$\underbrace{\underline{\underline{A}}^{-1}\underline{\underline{A}}}_{I}\underline{x} = \underline{\underline{A}}^{-1}\underline{b}$$

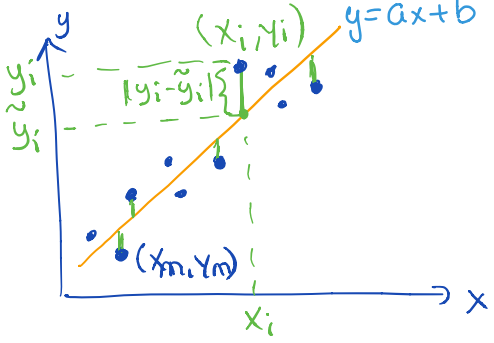
$$\boxed{\underline{x} = \underline{\underline{A}}^{-1}\underline{b}}$$

- Not very efficient if A is large
- Numerical and efficient methods exist

Overdetermined systems have more equations than unknowns

$[A]_{m \times n} \Rightarrow (m > n)$ → # columns = # unknowns; ↳ # rows = # equations

e.g. linear regression



"best" fitting line
(n=2 if 2D problem)
n coefficients is fit to m data pts. (x,y)

Underdetermined : less eq^{n}s than unknowns ($m < n$)

MATLAB : $\underline{\underline{A}}\ \underline{x} = \underline{b}$

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>> x = inv(A) * b
>> x = A \ b
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inverse of A

CHAPTER 9 : Gauss Elimination

Gauss elimination : combine eq^{n}s to eliminate unknown

Solving Small # of Eq^{n}s.

Graphical Method

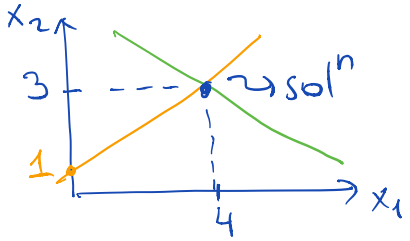
Ex

$$3x_1 + 2x_2 = 18$$
$$-x_1 + 2x_2 = 2$$

$$x_2 = -\frac{3}{2}x_1 + 9$$
$$x_2 = \frac{1}{2}x_1 + 1$$

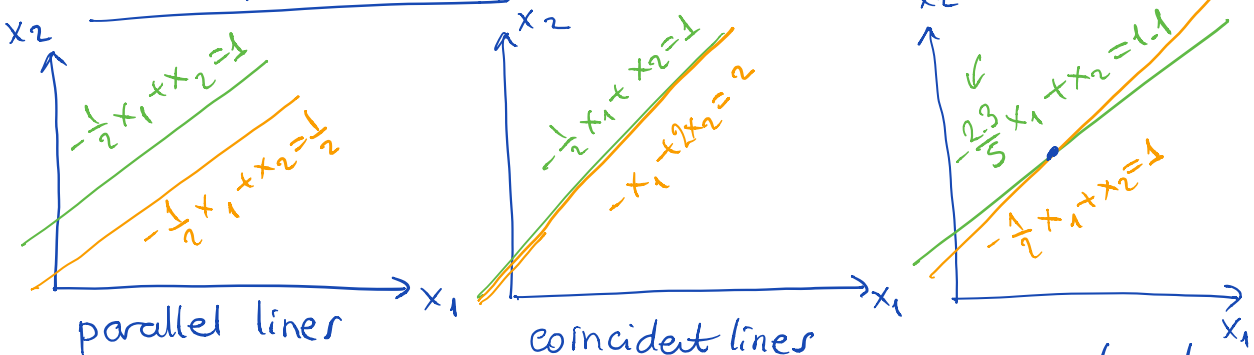
→ lines in 2D

slope, intercept



* 3 eqn in 3 unknowns ⇒ planes in 3D.
* Beyond 3 dims. graphical methods are not useful

Three problem cases:



parallel liner
no solution

coincident lines
infinite solutions

singular cases

very close to being singular
ill-conditioned
- difficult to identify the exact pt. at which they intersect